Contact outcomes across staffing and movement stochasticity

Study setup

This completed CPU experiment contains 2,100 successful simulations: seven HCW counts (2, 4, 6, 8, 16, 24 and 32), five numeric stochasticity levels (0, 0.25, 0.5, 0.75 and 1), two movement strategies and 30 seeded replicates per combination. Every combination has exactly 30 unique replicates; there are no failure records. All combinations were rerun, without importing historical results.

The synthetic layout retains four patients, a small shared-resource pool including three staff-room workstations, a 06:55–19:05 observation window, a one-second timestep and a 0.1-metre contact threshold. The schedule generator seed is 0. Replicate r uses seed 20261021 + r for r = 0,…,29. Fixed movement uses 0.01 coordinate units per timestep; deadline-paced movement uses remaining distance divided by time to the scheduled task start when that time is positive. Strategy differences also include dispatch/task-insertion behaviour. Neither strategy is a calibrated clinical walking model.

The run used 12 CPU workers and reduced scheduling priority. Parent elapsed time was approximately 5 hours 41 minutes including automatic post-processing. This is a public toy scenario, not the planned year-long UCLH study. No sensitive clinical records are included.

Measurement and uncertainty

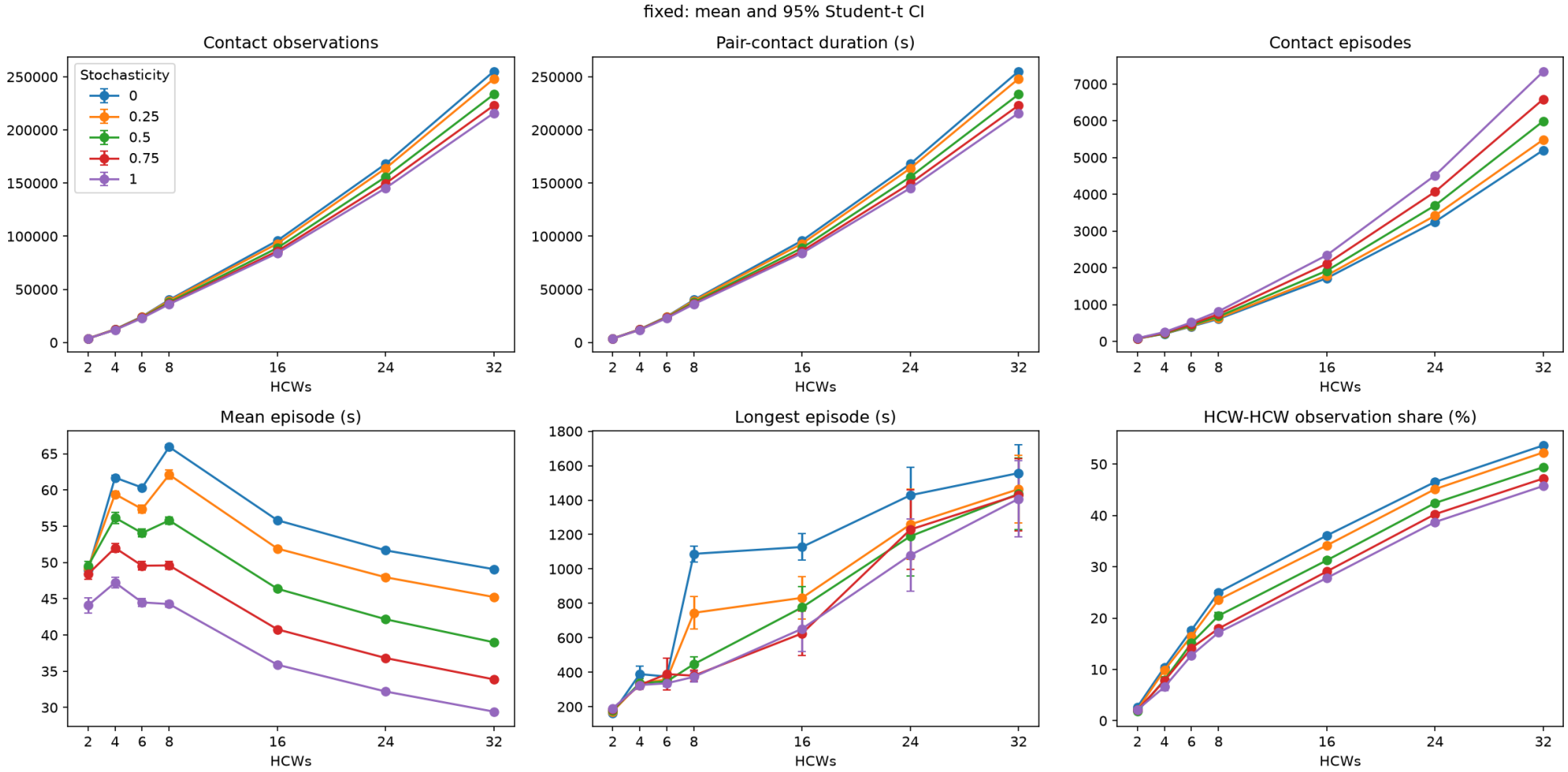
Contact observations count pairs within the threshold at recorded timesteps, grouped by building and floor and excluding pairs in different known rooms. Unknown room classifications can still contribute. Episodes join consecutive observations. Duration is the number of observations multiplied by the sample interval; initial and terminal samples are both recorded, so the interpretation of terminal-sample support requires care.

Total duration sums overlapping contacts across pairs: it is pair-contact time, not elapsed time on the ward. At this fixed one-second interval, observation counts and duration in seconds are numerically identical. More generally they are proportional at a constant interval; their pair-type shares remain identical. They are not independent evidence.

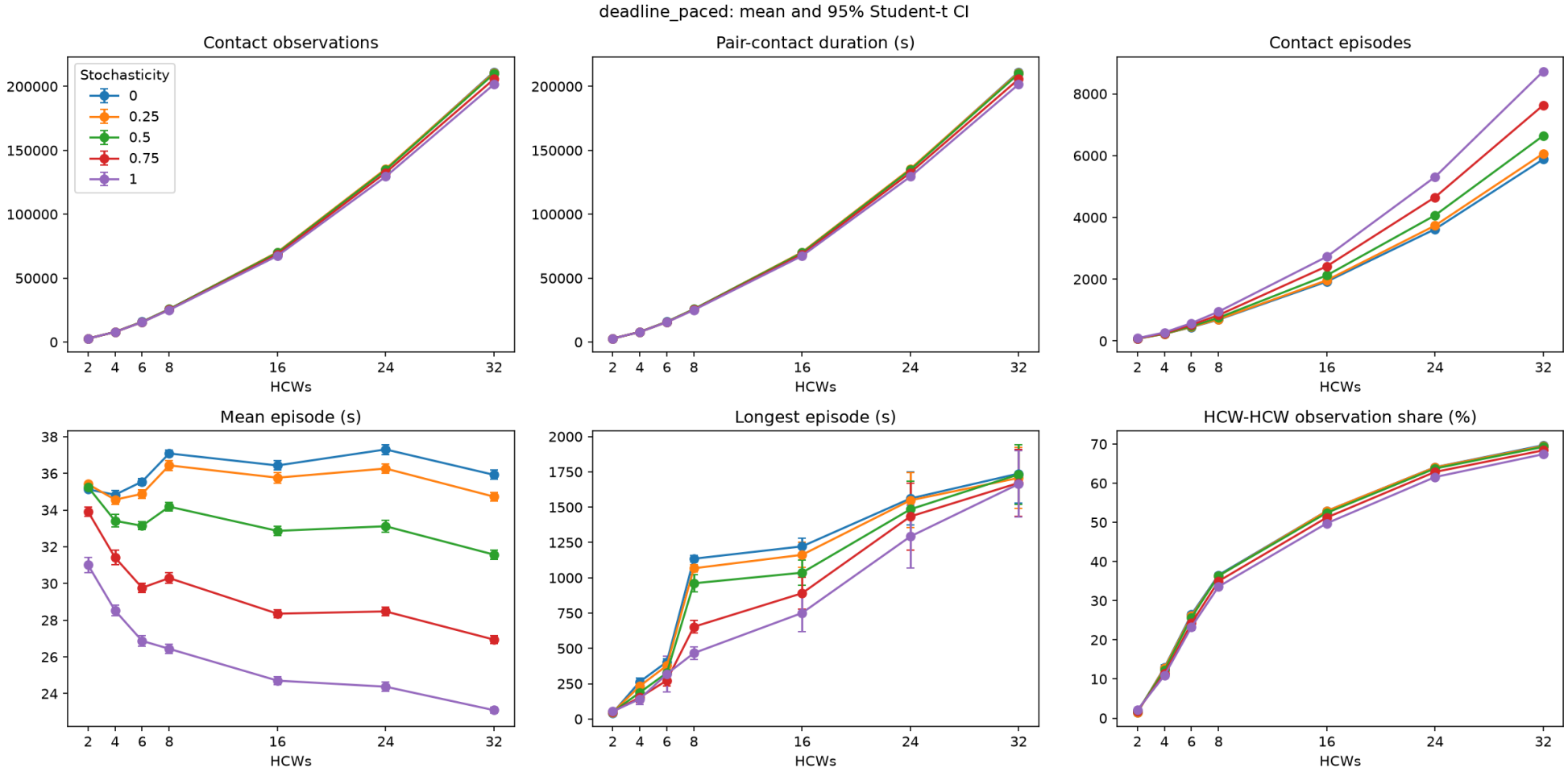
Tables report means [lower, upper 95% Student-t confidence limits] across 30 runs. For each metric the interval is mean ± t(0.975, n−1) × sample SD / sqrt(n). Mean episode length is computed within each run as total duration divided by episode count, then averaged across runs. Longest episode is the maximum within each run, then averaged. Pair-type shares are percentages computed within each run before averaging, not ratios of pooled totals. Undefined zero-denominator metrics are omitted; patient-patient episode length is undefined throughout, rather than zero. All overall metrics have n=30. Intervals describe seed variability conditional on this layout and schedule, and are unadjusted for multiple comparisons.

Staffing comparison

Contact observations increase with staffing at every stochasticity level for both strategies. Deadline-paced movement has fewer mean observations and a shorter mean episode than fixed movement in all 35 matched staffing/level cells. These are descriptive contrasts; confidence intervals for paired differences were not estimated here. Fewer proximity observations do not establish reduced infections or better care delivery.



Fixed strategy: six staffing panels with 95% confidence intervals



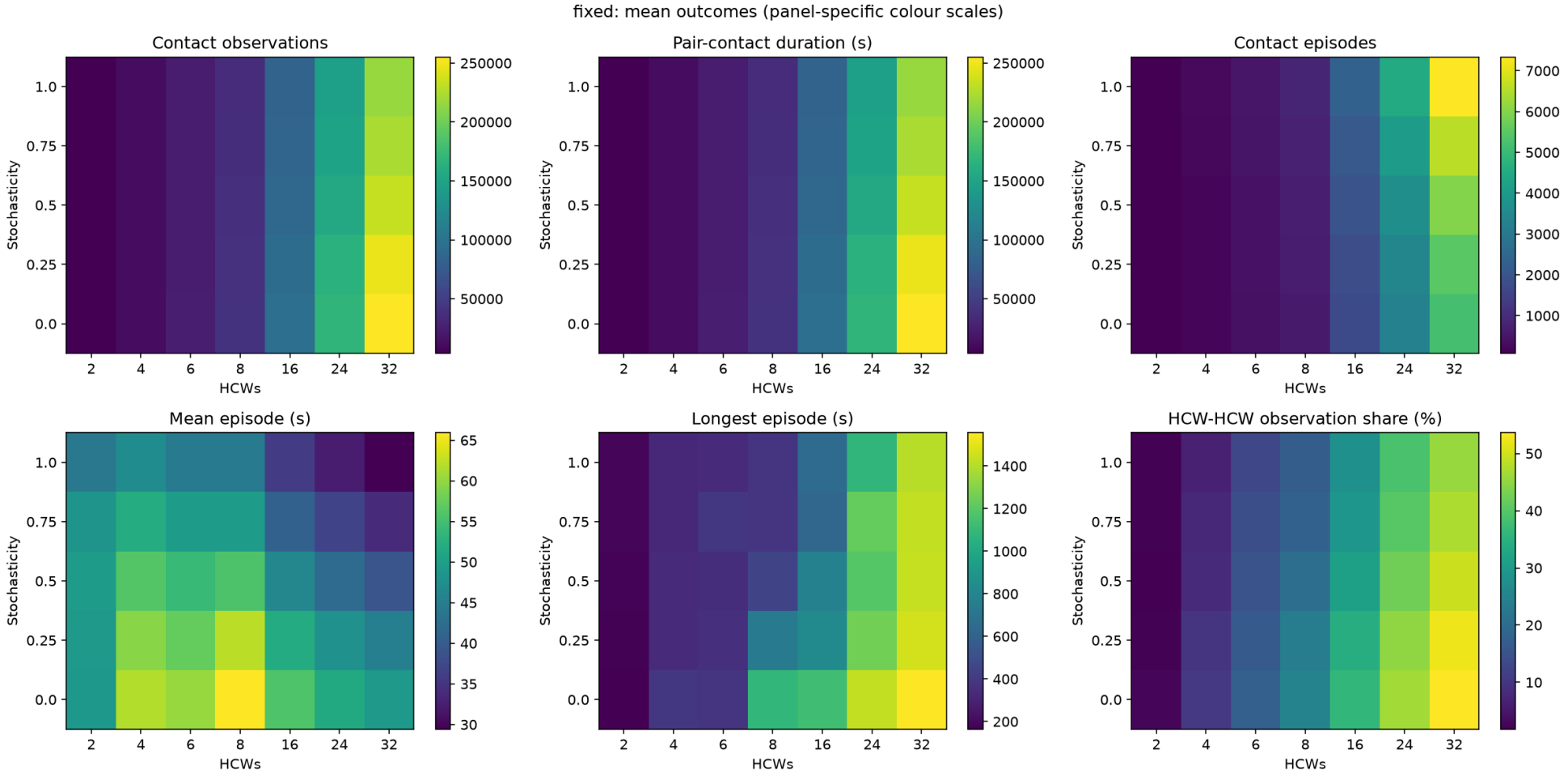
Deadline-paced strategy: six staffing panels with 95% confidence intervals

Numeric stochasticity response

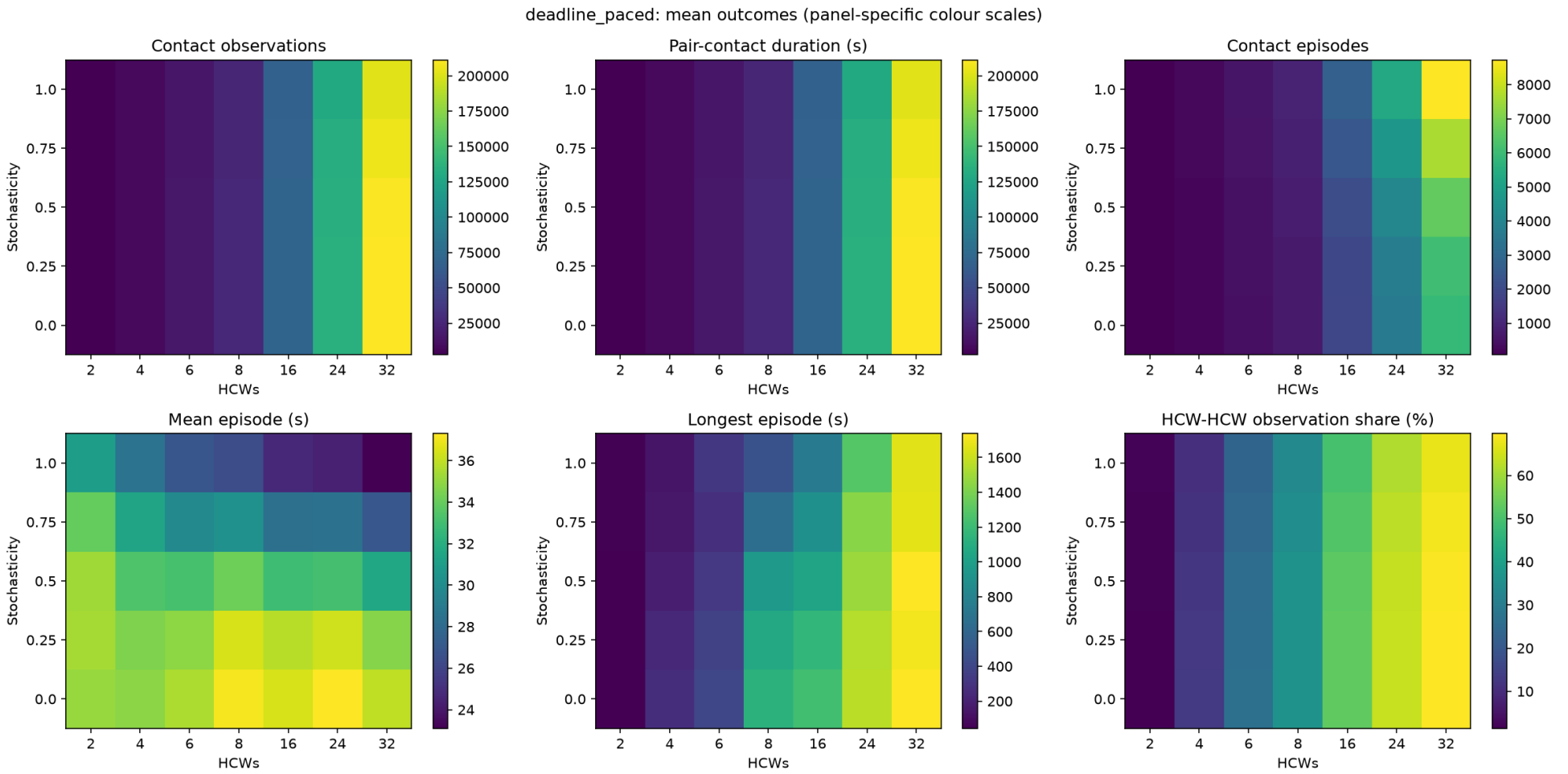
Increasing stochasticity from 0 to 1 reduces mean contact observations at every staffing level above 2 HCWs, for both strategies. At 2 HCWs the endpoint comparison instead increases observations. Fixed movement at 2 HCWs initially declines through 0.5 then rises; deadline-paced observations at 8, 16 and 24 HCWs also show small intermediate increases. The response is therefore not universally monotonic.

At 32 HCWs, fixed-strategy observations fall from 255,267.7 to 215,967.9 (15.4%), while deadline-paced observations fall from 211,303.1 to 201,615.8 (4.6%). Episodes increase from 5,202.6 to 7,340.6 for fixed and from 5,882.7 to 8,729.1 for deadline-paced movement. Mean episode lengths decrease from 49.1 to 29.4 seconds and from 35.9 to 23.1 seconds respectively. More episodes alongside less total pair-contact time is consistent with fragmentation of encounters, but this aggregate analysis does not identify the underlying trajectory-level cause.

The longest-episode response is less regular. At 6 HCWs with deadline pacing, its mean falls through level 0.75 and then rises at level 1. At 32 HCWs, deadline pacing produces longer mean run-maximum episodes than fixed movement at every level, despite lower total contact time. Total contact burden and extremes therefore do not rank strategies identically.



Fixed strategy heatmaps across staffing and numeric noise



Deadline-paced strategy heatmaps across staffing and numeric noise

Endpoint tables

Each table shows all five numeric levels. The complete seven-count results, including pair-type metrics, are supplied in the accompanying CSVs. Observations also equal pair-contact seconds in this experiment.

2 HCWs

Strategy	Level	Observations / pair-seconds	Episodes	Mean episode (s)	Longest episode (s)
Fixed	0	3,720.5 [3,719.9, 3,721.1]	76.0 [76.0, 76.0]	49.0 [48.9, 49.0]	164.0 [164.0, 164.0]
Fixed	0.25	3,709.9 [3,699.5, 3,720.3]	75.4 [74.8, 76.1]	49.2 [48.8, 49.6]	172.2 [171.0, 173.3]
Fixed	0.5	3,700.2 [3,680.8, 3,719.6]	74.8 [73.8, 75.9]	49.5 [48.9, 50.1]	177.9 [175.5, 180.3]
Fixed	0.75	3,739.2 [3,717.4, 3,761.0]	77.4 [76.2, 78.6]	48.4 [47.7, 49.1]	187.5 [184.7, 190.2]
Fixed	1	3,751.4 [3,716.3, 3,786.5]	85.4 [83.4, 87.4]	44.1 [43.0, 45.1]	190.7 [187.8, 193.7]
Deadline	0	2,689.7 [2,687.8, 2,691.5]	76.6 [76.4, 76.8]	35.1 [35.1, 35.2]	43.0 [43.0, 43.0]
Deadline	0.25	2,716.1 [2,711.8, 2,720.4]	76.7 [76.5, 76.9]	35.4 [35.3, 35.5]	47.2 [46.9, 47.5]
Deadline	0.5	2,727.3 [2,720.6, 2,734.0]	77.4 [76.9, 77.8]	35.3 [35.1, 35.4]	49.3 [48.6, 50.1]
Deadline	0.75	2,731.8 [2,720.5, 2,743.1]	80.6 [79.9, 81.2]	33.9 [33.7, 34.2]	51.3 [50.5, 52.0]
Deadline	1	2,747.5 [2,727.1, 2,767.8]	88.7 [87.3, 90.1]	31.0 [30.6, 31.4]	53.9 [52.9, 55.0]

6 HCWs

Strategy	Level	Observations / pair-seconds	Episodes	Mean episode (s)	Longest episode (s)
Fixed	0	24,480.1 [24,362.2, 24,598.1]	405.8 [403.9, 407.8]	60.3 [60.0, 60.6]	375.0 [351.7, 398.3]
Fixed	0.25	24,138.3 [23,989.1, 24,287.5]	420.9 [417.6, 424.3]	57.4 [56.8, 57.9]	359.0 [332.0, 385.9]
Fixed	0.5	23,732.4 [23,561.0, 23,903.9]	439.3 [434.3, 444.3]	54.1 [53.5, 54.6]	346.8 [322.0, 371.6]
Fixed	0.75	23,394.8 [23,234.9, 23,554.7]	472.6 [466.4, 478.8]	49.5 [48.9, 50.2]	389.6 [297.9, 481.2]
Fixed	1	22,990.8 [22,794.6, 23,187.1]	517.2 [510.3, 524.1]	44.5 [44.0, 45.0]	336.2 [315.3, 357.1]
Deadline	0	15,874.5 [15,779.5, 15,969.5]	446.5 [444.6, 448.5]	35.6 [35.4, 35.7]	404.3 [384.4, 424.3]
Deadline	0.25	15,835.3 [15,738.2, 15,932.4]	454.0 [451.2, 456.9]	34.9 [34.6, 35.1]	378.1 [353.4, 402.8]
Deadline	0.5	15,803.2 [15,708.4, 15,897.9]	476.9 [474.1, 479.7]	33.1 [32.9, 33.4]	326.4 [288.0, 364.7]
Deadline	0.75	15,570.9 [15,474.7, 15,667.1]	523.3 [518.1, 528.5]	29.8 [29.5, 30.0]	273.0 [234.8, 311.2]
Deadline	1	15,540.8 [15,345.5, 15,736.2]	578.5 [572.0, 585.0]	26.9 [26.6, 27.2]	318.6 [192.9, 444.3]

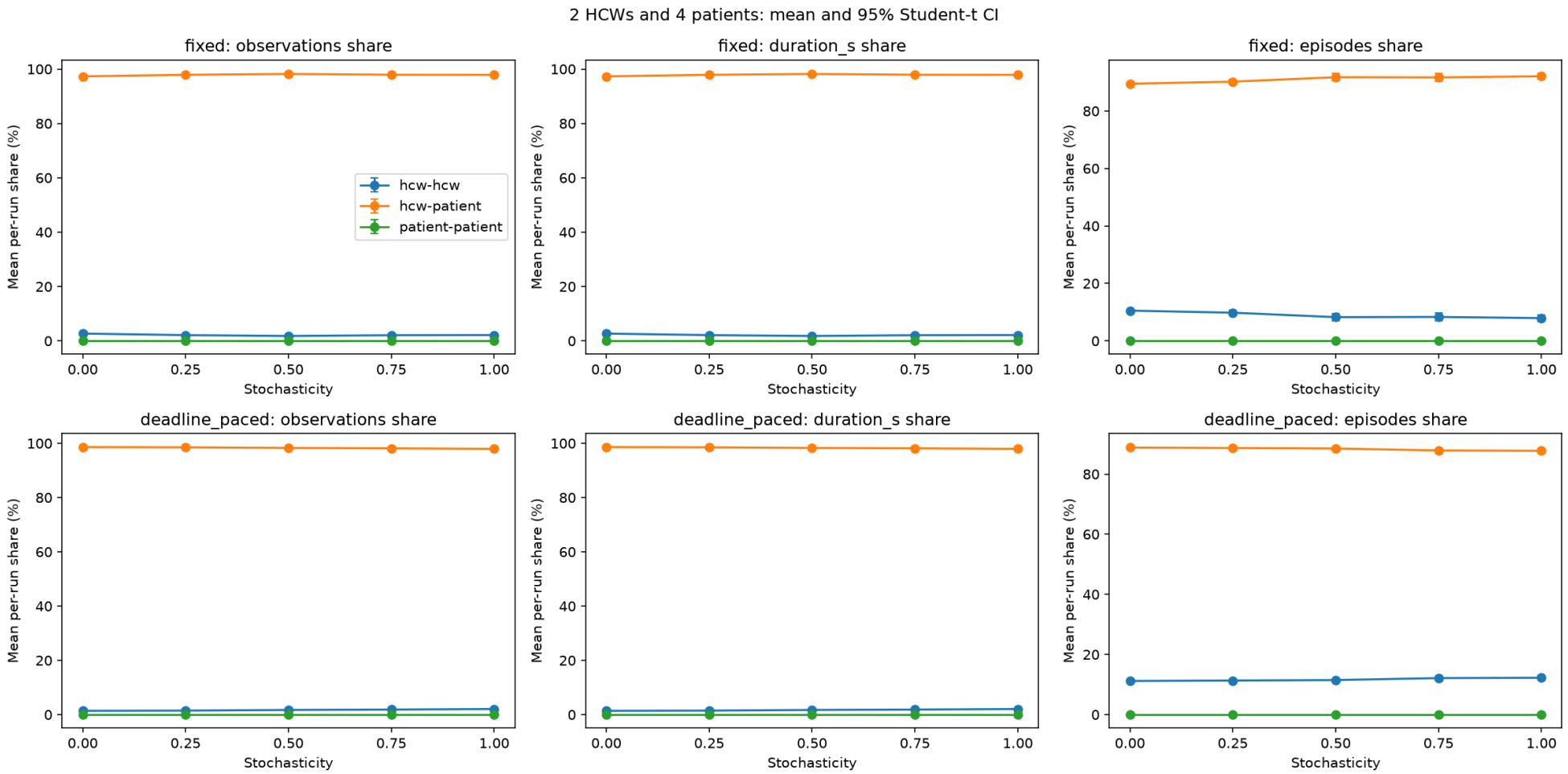
32 HCWs

Strategy	Level	Observations / pair-seconds	Episodes	Mean episode (s)	Longest episode (s)
Fixed	0	255,267.7 [254,167.1, 256,368.2]	5,202.6 [5,188.6, 5,216.6]	49.1 [48.8, 49.3]	1,556.7 [1,390.1, 1,723.3]
Fixed	0.25	248,208.4 [247,042.3, 249,374.4]	5,488.0 [5,465.9, 5,510.1]	45.2 [45.0, 45.5]	1,463.6 [1,266.8, 1,660.4]
Fixed	0.5	233,534.4 [232,406.2, 234,662.5]	5,990.8 [5,964.0, 6,017.7]	39.0 [38.8, 39.2]	1,436.0 [1,227.7, 1,644.2]
Fixed	0.75	223,182.9 [221,875.7, 224,490.1]	6,589.5 [6,556.1, 6,622.9]	33.9 [33.7, 34.1]	1,430.7 [1,221.3, 1,640.1]
Fixed	1	215,967.9 [214,685.4, 217,250.4]	7,340.6 [7,300.3, 7,381.0]	29.4 [29.2, 29.6]	1,407.7 [1,187.3, 1,628.0]
Deadline	0	211,303.1 [209,843.5, 212,762.7]	5,882.7 [5,864.9, 5,900.5]	35.9 [35.7, 36.2]	1,737.8 [1,531.2, 1,944.4]
Deadline	0.25	210,720.5 [209,124.2, 212,316.9]	6,067.2 [6,049.2, 6,085.3]	34.7 [34.5, 35.0]	1,707.0 [1,491.1, 1,923.0]
Deadline	0.5	209,797.5 [208,047.4, 211,547.5]	6,644.9 [6,620.7, 6,669.1]	31.6 [31.3, 31.8]	1,731.7 [1,520.6, 1,942.8]
Deadline	0.75	205,642.6 [203,925.2, 207,360.0]	7,633.1 [7,605.1, 7,661.1]	26.9 [26.7, 27.2]	1,671.5 [1,433.4, 1,909.5]
Deadline	1	201,615.8 [200,091.7, 203,140.0]	8,729.1 [8,689.4, 8,768.7]	23.1 [22.9, 23.2]	1,663.8 [1,428.5, 1,899.0]

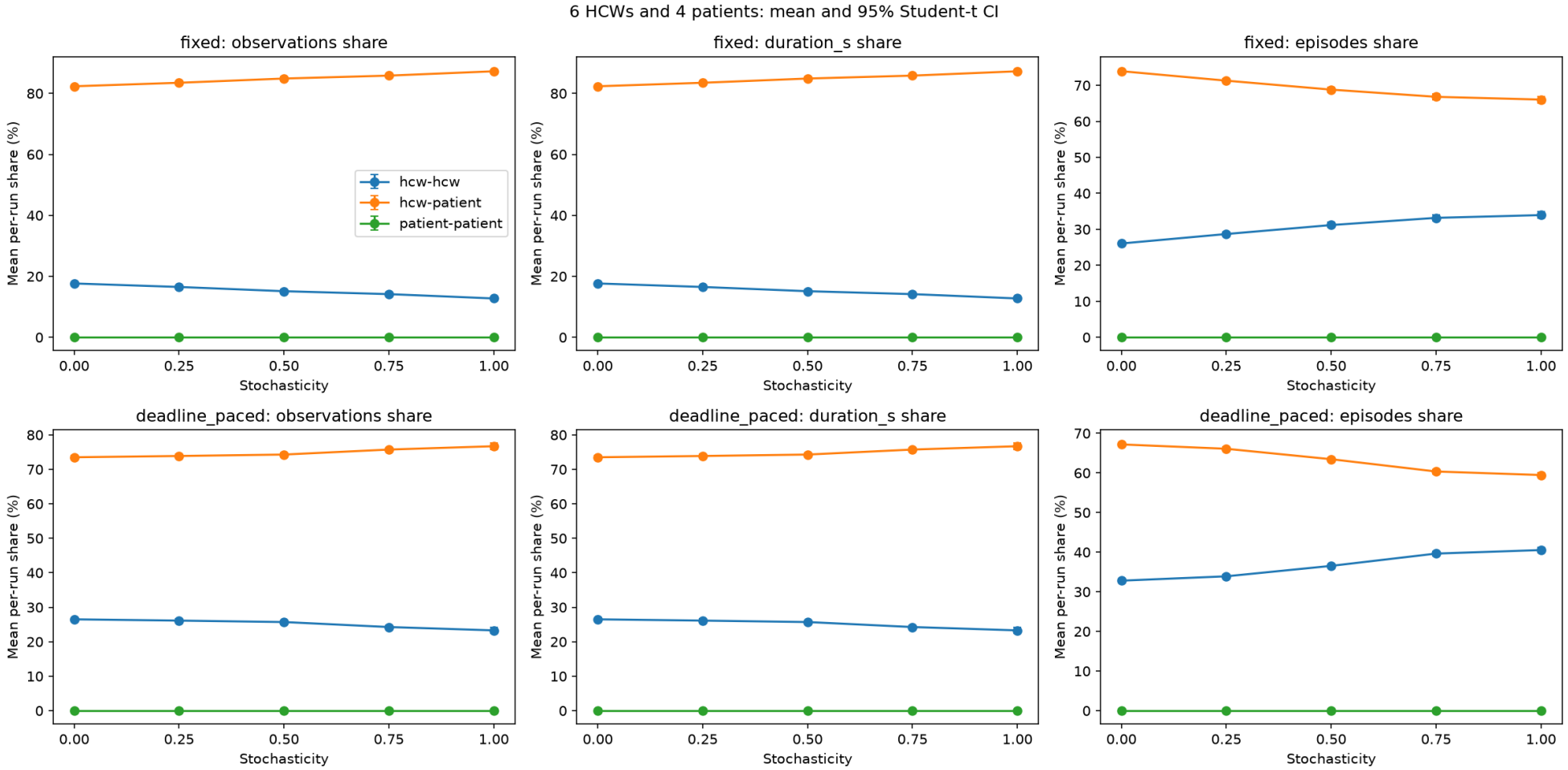
Pair-type comparisons

Patient-patient observations and episodes are zero in all 2,100 runs. This is an outcome of this fixture and detector, not evidence that patient-patient transmission is absent in hospitals. The raw results verify absence under the detector; within-patient-type mean episode lengths are undefined.

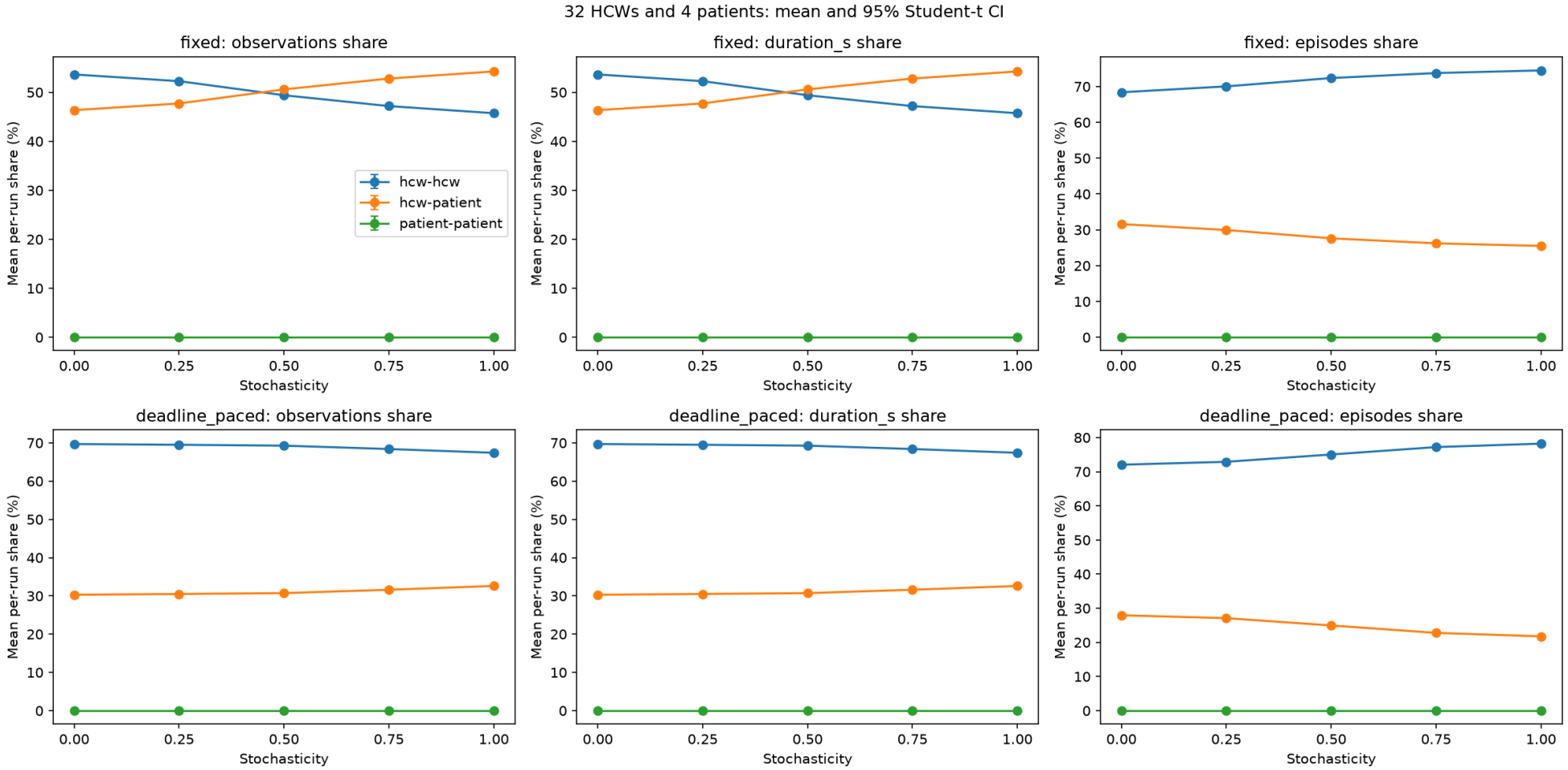
At 2 HCWs, HCW-patient encounters account for approximately 97–99% of observations. At 32 HCWs, the mean HCW-HCW share ranges from 45.7–53.6% for fixed movement and 67.4–69.7% for deadline pacing. A larger share is not necessarily a larger absolute count: the denominators differ by strategy. Increasing HCWs also changes possible pair counts: $H(H-1)/2$ HCW-HCW pairs versus $4H$ HCW-patient pairs. Changing shares cannot be interpreted solely as altered behaviour.



Pair-type comparisons at 2 HCWs



Pair-type comparisons at 6 HCWs



Pair-type comparisons at 32 HCWs

Coupled noise mechanism

For numeric stochasticity s, each CPU movement proposal adds a normal heading perturbation with standard deviation s degrees. It then independently multiplies the x and y displacement components by 1 + Normal(0, s). The parameter therefore couples small angular noise and dimensionless displacement noise; this is not a pure heading-noise experiment. The implementation was preserved for the sweep.

For s > 0, each component multiplier is negative with probability Phi(−1/s). At s = 0.25, 0.5, 0.75 and 1 these probabilities are approximately 0.0032%, 2.28%, 9.12% and 15.87%. They describe proposals, not observed accepted-step reversal rates; a zero displacement component does not move merely because its multiplier is negative. Independent component multipliers also make noise depend on coordinate orientation.

Movement checks retry proposals up to five times for room/wall constraints. The implementation returns its final proposal even if all attempts fail. Collision-safe movement is therefore not guaranteed. This report does not quantify invalid-position frequency or attribute a measured effect to it. Any correction should produce a separately versioned experiment, not silently replace these outcomes.

At s = 0, placement, task durations and other random processes can still vary with seed. Reusing the same seed across conditions does not guarantee identical random-number consumption after trajectories diverge.

Interpretation and limitations

The principal finding is that staffing, movement strategy and this coupled noise parameter jointly alter contact totals, episode fragmentation and pair composition in the synthetic fixture. The reduction in total contact time with higher noise at larger staffing does not apply uniformly to every metric or staffing level.

This experiment uses one layout and one generated schedule per staffing count. Adding HCWs changes activity volume and potential pairs while keeping patients and shared resources fixed; it is not a fixed-care-demand staffing intervention. Seed intervals do not cover layout, roster, observation or structural uncertainty. Longest episodes are extreme statistics whose distributions deserve further inspection.

The CPU experiment measures proximity, not colonisation, infection or antimicrobial resistance acquisition. No visit-completion or care-quality measurements were collected in this sweep. Results therefore support neither a safety ranking nor a clinical staffing recommendation. The 0.1-metre point-agent threshold, uncalibrated speed, unresolved-room handling and sample-count duration convention limit physical interpretation.

The next analysis should measure movement invariants, care completion and speed distributions, separate angular and displacement noise, and compare fixed-workload and resource-scaled scenarios. Those are proposed follow-up experiments, not completed validations.

Reproducibility and validation

The seven figures are unchanged copies of the completed sweep outputs. The accompanying raw results, complete summary, overall summary and pair-type summary are included for review. The original experiment manifest preserves design and code/input SHA-256 fingerprints; its paths refer to the original sweep worktree. This documentation branch does not contain the uncommitted runner changes or large execution checkpoints from that worktree.

Validation recomputed summary means and Student-t intervals from all 2,100 raw rows and compared every exported summary entry. It also checked the complete factorial design, seed schedule, unique identifiers and zero patient-patient observations. Table cells are generated directly from summary.csv. Figure and data hashes, PDF rendering version and inspection results are recorded in validation.json. Original reports and figures remain unchanged.